



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2025-2026

Digital Learning Handout

Part A: Content Design

Course Title	Data Management for Machine Learning
Course No(s)	DSECL ZG529 / AIMLC ZG529
Credit Units	4
Credit Model	3-1-0 (XX Hours of Class-room Instruction + XX Hours of Case-studies/Tutorials/Laboratories + XX Hours of Student Preparation)
Instructors	Prof Sandeep Patil, Prof Sunil Bhutada, Prof D VENKATA SUBRAMANIAN
Version No:	1.3
Date:	20 th October 2025

Course Description:

Data Models and Query Languages: Relational, Object-Relational, NoSQL data models; Declarative (SQL) and Imperative (MapReduce) Querying; Data Encoding: Evolution, Formats, Models of dataflow; Machine learning workflow; Data management challenges in ML workflow; Data Pipelines and patterns; Data Pipeline Stages: Data extraction, ingestion, cleaning, wrangling, versioning, transformation, exploration, feature management; Modern Data Infrastructure: Diverse data sources, Cloud data warehouses and lakes, Data Ingestion tools, Data transformation and modelling tools, Workflow orchestration platforms; ML model metadata and Registry, ML Observability, Data privacy and anonymity.

Course Objectives

No	Course Objective
CO1	Introduce students to the data models, storages and querying languages used in data management emphasizing on machine learning aspects
CO2	Provide required guidance on architecture of modern data platform, usage and types of data pipelines
CO3	Deliver Hands-on exposure to the common techniques, and tools used by data engineers to support build, test, deploy and automate the machine learning pipelines
CO4	Expose students to the industry best practices essential to deal with data privacy, metadata and observability

Text Book(s):

T1	Fundamentals of Data Engineering: Plan and Build Robust Data Systems by Reis and Housley
T2	Reliable Machine Learning by Cathy Chen, Niall Richard Murphy, Kranti Parisa, D. Sculley, Todd Underwood

Reference Book(s) & other resources:

R1	Designing Data-Intensive Applications by Martin Kleppmann
R2	Data Pipelines Pocket Reference by Densore





R3	Building Machine Learning Pipelines by Hapke, Nelson
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Learning Outcomes: Students will be able to

LO1	Explain and differentiate the necessity, position and role of data management components appearing in the modern data stacks
LO2	Analyse and evaluate data ingestion, flow, and processing challenges and propose suitable architectural solutions.
LO3	Design and implement dataflows for machine learning pipelines using contemporary data engineering tools
LO4	Apply the acquired conceptual data management knowledge and practices over a real-world machine learning workflow addressing the model metadata, privacy and monitoring aspects

Modular Content Structure

M1 Foundations of data management

- Data Representations
- Data Management Fundamentals
- Data Architectures

M2 Modern Data Platform

- Data Pipelines
- Modern Data Infrastructure and DataOps
- ML Workflow/lifecycle

M3 Data Management in ML Workflow

- Data Ingestion
- Data Profiling and Validation
- Analytics Engineering
- Workflow Orchestration and Automation
- Feature Preparations
- ML Experimentation and Metadata

M4 Frontier Practices in Data & ML Systems

- Distributed and Scalable Data Processing
- Data Pipelines for LLMs
- Data Privacy, Governance and Responsible ML
- Data and ML Observability

Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	Data Representations	• Data Formats • Data Models and Query Languages	R1 Chapter 2,3,4





		• Data Encoding / Serializations • Data Processing	
2	Data Management Fundamentals	• Data Management Principles • Data Management Components • Data Management Roles • Phases of Data	T2 chapter 2
3	Data Architectures	• Data Architectures – Defined • Types of Data Architecture: Traditional vs Modern • Emerging Data Architectures • Lakehouse vs Data Warehouse vs Data Lake • Generalized Architecture of Big Data Systems	T1 Chapter 3,4 R2 Chapter 2,3
4	Data Pipelines	• Modes of Data Flow • Modern Data Stack • Data Pipelines • ETL, ELT • Unified data flow • Tools / Platforms	T1 Chapter 6
5	Modern Data Infrastructure and DataOps	• Modern Data Infrastructure • Data Science Infrastructure • Data Serving for Analytics and ML • Automation in Data Infrastructure Management • DevOps Foundation • Continuous Training (CT) and Continuous Deployment (CD) pipelines • Intro to DataOps and Observability • Case Study: Modern ML Data Infrastructure (integrating DataOps & Observability)	T1 Chapter 6
6	ML lifecycle and Workflow	• ML Lifecycle • Three Levels of ML Software • Data: Data Engineering Pipelines	T2 Chapter 1 R3 Chapter 1





		- Model: Machine Learning Pipelines - Different forms of ML workflows - Code: Deployment Pipelines - DataOps → MLOps relationship - Case survey: model deployment pipeline failures	
7	Data Collection and Ingestion	- Data Sources - Data Generation in Source Systems - Data Ingestion - Data contracts, Lineage and Governance - Data Ingestion Considerations - Ways to Ingest Data	T1 Chapter 5, 7
8	Data Profiling and Validation	- Common Problems with Data - Data Quality Dimensions - Bias and Fairness - Data Leakage - Data Skew and Drift - Drift detection metrics - Profiling for ML readiness - Filter / Selection / Sampling, Validation - Data Validation Approaches and Frameworks - Automated Data Quality Testing in Pipelines (Great Expectations, Soda Core, Deequ) - pre-ingestion validation vs pre-training validation	R3 Chapter 4
9	Analytics Engineering	- Data Integration - Transformation and Modeling Patterns (ELT, dbt-style) - Testing in Data Transformations - Data Transformation - Data Transformation with dbt - Data Versioning - Data Lineage and Reproducibility - Documentation of Data Models (dbt docs, DataHub, Amundsen)	Classroom discussions Demonstrations





10	Workflow Orchestration and Automation	• Need for Orchestration • Data Pipeline vs Workflow • Core Concepts • Common Orchestration Tools - Airflow, Prefect, Dagster overview. • Designing an Orchestrated Workflow • Integrating DataOps with Orchestration • Best Practices	Classroom discussions Demonstrations
11	Feature Preparations	• Feature life cycle • Data Annotation / labelling • Data augmentation and Data Synthesis • Common Feature Engineering Operations • Feature Importance • Feature Generalization • Feature Reuse and Sharing Practices • Feature Stores • Feature Governance • Engineering Good Features	T2 Chapter 4
12	ML Experimentation & Metadata	• Model training & experimentation • Model Analysis & Validation • ML Metadata Store • Dataset, Feature, Label, Pipeline metadata • ML Experiment Tracking • Experiment tracking frameworks (MLflow, Neptune, Weights & Biases) • Model Registry and Versioning • Metadata governance and lineage tracking	Classroom discussions
13	Distributed & Scalable Data Processing	• Big Data Analytics landscape • Distributed and Parallel data processing principles • In-memory and stream processing paradigms	Classroom discussions Demonstrations





		• Modern distributed frameworks – Spark Structured Streaming, Flink, Ray, Dask • Cloud-native data platforms (Snowflake, BigQuery, Databricks) • Batch vs. Streaming trade-offs for ML training vs inference pipeline	
14	Data Pipelines for LLMs	• Why LLM Data Pipelines are Different? • Types of LLM Pipelines • Core Components of an LLM Data Pipeline • Data Pipeline Architecture for LLM Systems • Key Engineering Considerations • RAG pipeline frameworks: LangChain, LlamaIndex, Haystack	Classroom discussions Demonstrations
15	Data Privacy, Governance & Responsible ML	• Data Privacy Landscape and Regulations • Governance frameworks: DPDP Act (India), GDPR, CCPA — global compliance perspective • Privacy-Preserving ML Paradigms (Differential Privacy, Federated Learning) • Anonymization and Synthetic Data • Encrypted ML and Secure Computation	T2 Chapter 6 R3 Chapter 14
16	Data and ML Observability	• Why Observability Matters? • Five Pillars of Data Observability • Practical Implementation • Connecting to DataOps • From Data to ML Observability • Common ML System Failures • Monitoring Metrics • Tooling Landscape • Unified Observability Framework	T2 Chapter 9



Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc..) number of lab exercises needed, etc.

1. Lab work: 04
2. Project work: 01
3. Case Study: 02
4. Simulation: 0
5. Work Integrated Learning Assignment: 0
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

- Design and implement data flows using databases, REST/RPC, and message brokers
- Build a modern data stack with ELT pipelines, cloud storage, transformation tools, and BI platforms
- Manage machine learning model metadata using MLFlow or Neptune for tracking and reproducibility
- Construct a machine learning pipeline integrating data versioning, feature stores, and prediction services

Scope of Experiential Learning Component:

- Identification and usage of various tools / frameworks available for building an end-to-end data plus machine learning pipeline

Lab Infrastructure:

- CSIS Remote Labs / Local setups

List of Experiments:

Exp No.	Experiment Title	Reference to handout module/session
1.	Design and implement the simple data flows involving various data formats Modes of data flows • Through Databases – use SQL / Custom Program to read/write into databases • Through REST/RPC – Synchronous mechanism for data exchange	M1





	Through Message Brokers / Queues – Asynchronous mechanism for data exchange	
2.	Build a Modern Data Stack Components - a fully managed ELT data pipeline - a cloud-based columnar warehouse or data lake as a destination - a data transformation tool - A business intelligence or data visualization platform.	M2
3.	Manage Machine Learning Model Metadata using MLFlow / Neptune Components - Projects - Experiments - Model metadata - Model tracking / logging - Model Registry	M3
4.	Construct a Machine Learning Pipeline with Data Versioning Tool Components - Data Pipeline - Data Versioning Tool - Feature Store - ML Pipeline - Prediction Service	M3

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name	Type (Open book, Closed book, Online,	Weight	Duration	Day, Date, Session, Time
	(Quiz, Lab, Project, Mid-term				





	exam, End semester exam, etc.)	etc.)			
EC – 1	Assignment 1	Take Home / Online	20%	21 days	TBA
	Quiz 1		10%	7 days	
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	20-06-2026 (EN)
EC - 3	Comprehensive Exam	Open Book	40%	2½ Hours	05-09-2026 (EN)

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.
[EC3 Guidelines](#)
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

